

Task-relevant information acquisition

PART I - INFORMATION CHARACTERIZATION

a

Complete sensing field

Information-rich field
!= engineering decision



b

Limited sensing under exact cost

Only a legal subset
can be measured



c

Task-relevant value

Candidates differ in
downstream CAI value



d

State-conditioned acquisition loop

$I_t \rightarrow \text{value} \rightarrow \text{legal action}$
 $\rightarrow \text{reveal} \rightarrow I_{t+1}$

Retrospective teacher/oracle evidence | characterization only

Deployable decision uses
legal state and exact cost only

legal-state boundary

WHY: complete-field information is not the same as decision-relevant measurement value